

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0688

Longview2 345 kV
September 20, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0688. The Interconnection Customer proposes a 536.7 MW solar photovoltaic facility interconnecting at the Longview2 345 kV substation in the West Virginia area. The request was received on October 13, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 336 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$51,473,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: ASPEN to VILLA MAR 2 1658 138kV
Contingency: ASPEN to CRANDALL 345kV line
Loading: 150.4% (Rating: 227 MVA)
DFAX: 5.8%
Pre-project loading: 136.7%
2. Facility: CARODEAN to REDLAND 138kV Line
Contingency: CRANDALL to DAYTON 345kV
Loading: 144.3% (Rating: 222 MVA)
DFAX: 6.6%
Pre-project loading: 128.4%
3. Facility: ORCHARD to VILLA MAR 2 1658 138kV
Contingency: CRANDALL to ASPEN 345kV
Loading: 131.2% (Rating: 249 MVA)
DFAX: 5.8%
Pre-project loading: 118.7%
4. Facility: WEYERS CAVE to GREENBANK 138kV
Contingency: CRANDALL to DAYTON 345kV Line
Loading: 130.4% (Rating: 265 MVA)
DFAX: 6.8%
Pre-project loading: 116.6%
5. Facility: DECLO to ORCHARD 138kV
Contingency: CRANDALL to ASPEN 345kV
Loading: 128.8% (Rating: 244 MVA)
DFAX: 5.8%
Pre-project loading: 116.0%

6. Facility: FRITO LAY to DPC PRAIRIE DU CHIEN SUBSTATION 138kV
Contingency: CHATM to PURDY 138kV
Loading: 125.9% (Rating: 329 MVA)
DFAX: 5.0%
Pre-project loading: 117.7%
7. Facility: HARRISONBURG to SANS SOUCI 138kV line
Contingency: DAYTON to CHATOM 345kV
Loading: 125.2% (Rating: 254 MVA)
DFAX: 5.0%
Pre-project loading: 114.6%
8. Facility: OAK CREEKSTA to SANS SOUCI 138kV
Contingency: CHATOM to DAYTON 345kV Line
Loading: 124.8% (Rating: 255 MVA)
DFAX: 5.0%
Pre-project loading: 114.2%
9. Facility: CARODEAN to CRANDALL 138kV
Contingency: DAYTON to CRANDLL 345kV
Loading: 122.4% (Rating: 284 MVA)
DFAX: 6.6%
Pre-project loading: 110.0%
10. Facility: WADSWORTH to OAK CREEK STATION 138kV
Contingency: DAYTN to CHATOM 345kV
Loading: 121.7% (Rating: 230 MVA)
DFAX: 5.0%
Pre-project loading: 109.9%
11. Facility: CHATOM to PURDY 138kV Line
Contingency: DPC PRAIRIE DU CHIEN to FRITO LAY 138kV
Loading: 119.8% (Rating: 421 MVA)
DFAX: 5.9%
Pre-project loading: 112.3%
12. Facility: WEYERS CAVE to GROTTUES 138kV Line
Contingency: DAYON SUBSTATION to CRANDALL 345kV
Loading: 111.3% (Rating: 417 MVA)
DFAX: 6.8%
Pre-project loading: 102.5%
13. Facility: SHERBURNE 345-138 kV Transformer
Contingency: SHERBURNE to PINETREE 345kV
Loading: 108.6% (Rating: 530 MVA)
DFAX: 15.3%
Pre-project loading: 93.0%
14. Facility: CRANDALL 345/138 kV Auto
Contingency: CRANALL to DAYTON 345kV L
Loading: 107.1% (Rating: 375 MVA)
DFAX: 6.6%
Pre-project loading: 97.6%
15. Facility: COMERIO 9703 to GREENLAND GAP 345kV L
Contingency: VENICE HLL to GREENLAND GAP 345kV Line
Loading: 106.2% (Rating: 1,272 MVA)
DFAX: 9.7%
Pre-project loading: 102.1%
16. Facility: SPINDLE to COMERIO 9703 345kV
Contingency: VENICE HILL to GREENLAND GAP 345kV L

Loading: 105.5% (Rating: 1,278 MVA)

DFAX: 9.7%

Pre-project loading: 101.4%

17. Facility: VENICE HILL to GREENLAND GAP 345kV

Contingency: SPINDLE to COMERIO 9703 345kV L

Loading: 105.2% (Rating: 1,285 MVA)

DFAX: 9.7%

Pre-project loading: 101.2%

18. Facility: SEMINOLE ELECTRIC COOPERATIVE to VAN DYKE 138kV

Contingency: PINETREE to SHERBURNE 345kV

Loading: 104.4% (Rating: 329 MVA)

DFAX: 9.6%

Pre-project loading: 88.8%

19. Facility: VENICE HILL to SPINDLE SUBSTATION 345kV L

Contingency: COMERIO 9703 to SPINDLE 345kV L

Loading: 104.3% (Rating: 1,280 MVA)

DFAX: 9.7%

Pre-project loading: 100.2%

20. Facility: MACKAY to SHERBURNE 138kV Line

Contingency: PINETREE to SHERBURNE 345kV

Loading: 104.0% (Rating: 337 MVA)

DFAX: 9.6%

Pre-project loading: 88.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VILLA MAR 21658 to ASPEN 138kV	Rebuild line with higher-capacity conductor	\$2,988,000	21% (\$635,000)
REDLAND to CARODEAN 138kV	Rebuild line with higher-capacity conductor	\$3,206,000	26% (\$846,000)
ORCHARD to VILLA MAR 2 1658 138kV	Rebuild line with higher-capacity conductor	\$7,669,000	21% (\$1,630,000)
WEYERS CAVE SUBSTATION to GREENBANK 138kV L	Rebuild line with higher-capacity conductor	\$3,511,000	24% (\$839,000)
DECLO to ORCHARD 138kV	Rebuild line with higher-capacity conductor	\$14,076,000	21% (\$2,993,000)
FRITO LAY to DPC PRAIRIE DU CHIEN 138kV	Rebuild line with higher-capacity conductor	\$3,267,000	41% (\$1,353,000)
HARRISONBURG to SANS SOUCI 138kV	Rebuild line with higher-capacity conductor	\$2,874,000	41% (\$1,167,000)
OAK CRK STATION to SANS SOUCI 138kV	Rebuild line with higher-capacity conductor	\$2,247,000	41% (\$912,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CARODEAN to CRANDALL SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$3,248,000	26% (\$858,000)
OAK CREEK STATION to WADSWORTH 138kV	Rebuild line with higher-capacity conductor	\$5,935,000	41% (\$2,410,000)
CHATOM to PURDY 138kV	Rebuild line with higher-capacity conductor	\$5,033,000	41% (\$2,068,000)
GROTTOES to WEYERS CAVE 138kV	Rebuild line with higher-capacity conductor	\$5,126,000	24% (\$1,209,000)
SERBURNE 345/138kV TX	Replace transformer with higher-rated unit	\$20,199,000	35% (\$7,095,000)
CRANDALL 345/138 kV Auto	Replace transformer with higher-rated unit	\$14,737,000	27% (\$3,978,000)
COMERIO 9703 to GREENLAND GAP 345kV L	Reconductor line	\$20,443,000	33% (\$6,823,000)
SPINDLE to COMERIO 9703 345kV	Reconductor line	\$18,574,000	34% (\$6,269,000)
VENICE HILL to GREENLAND GAP 345kV	Reconductor line	\$9,595,000	34% (\$3,234,000)
SEMINOLE ELECTRIC COOPERATIVE to VAN DYKE 138kV	Reconductor line	\$3,911,000	43% (\$1,681,000)
VENICE HILL to SPINDLE SUBSTATION 345kV L	Reconductor line	\$14,210,000	34% (\$4,789,000)
MACKAY to SHERBURNE 138kV Line	Reconductor line	\$1,590,000	43% (\$684,000)
	Total allocated to project		\$51,473,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.